

CDC Diabetes Health Indicators Analysis

Josie Charles, Ray Prasad, Dawson Ren

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Introduction

This project analyzes the CDC Diabetes Health Indicators dataset to predict whether an individual has diabetes based on various lifestyle factors and health indicators. The goal is to build a binary classification model that can accurately identify individuals at risk of diabetes. We will explore the data, handle multicollinearity, and compare different logistic regression models, including those with regularization (LASSO and Ridge).

Data Description

The dataset is obtained from the UCI Machine Learning Repository (ID: 891). It contains health indicators for 253,680 individuals.

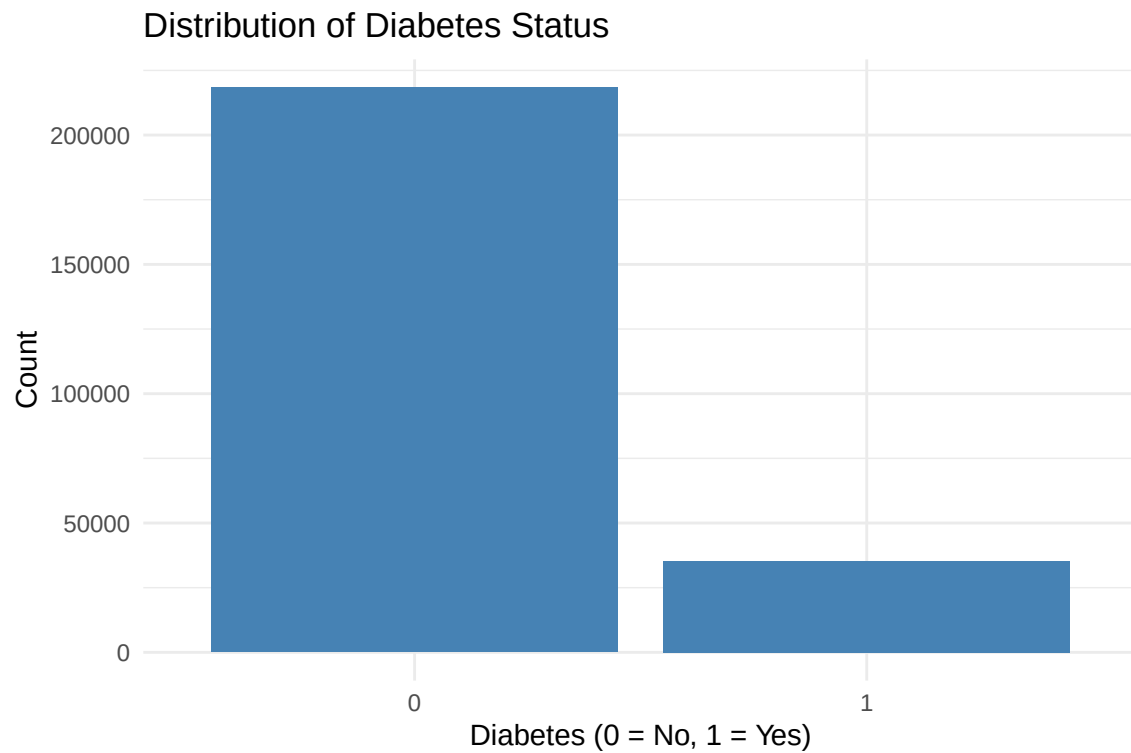
Raw Data and Pre-processing

The raw data was fetched using the `ucimlrepo` Python package and exported to a CSV file. The dataset includes a binary target variable `Diabetes_binary` (0 = no diabetes, 1 = pre-diabetes or diabetes) and 21 feature variables.

The dataset contains 253680 observations and 22 variables. There are no missing values in the dataset.

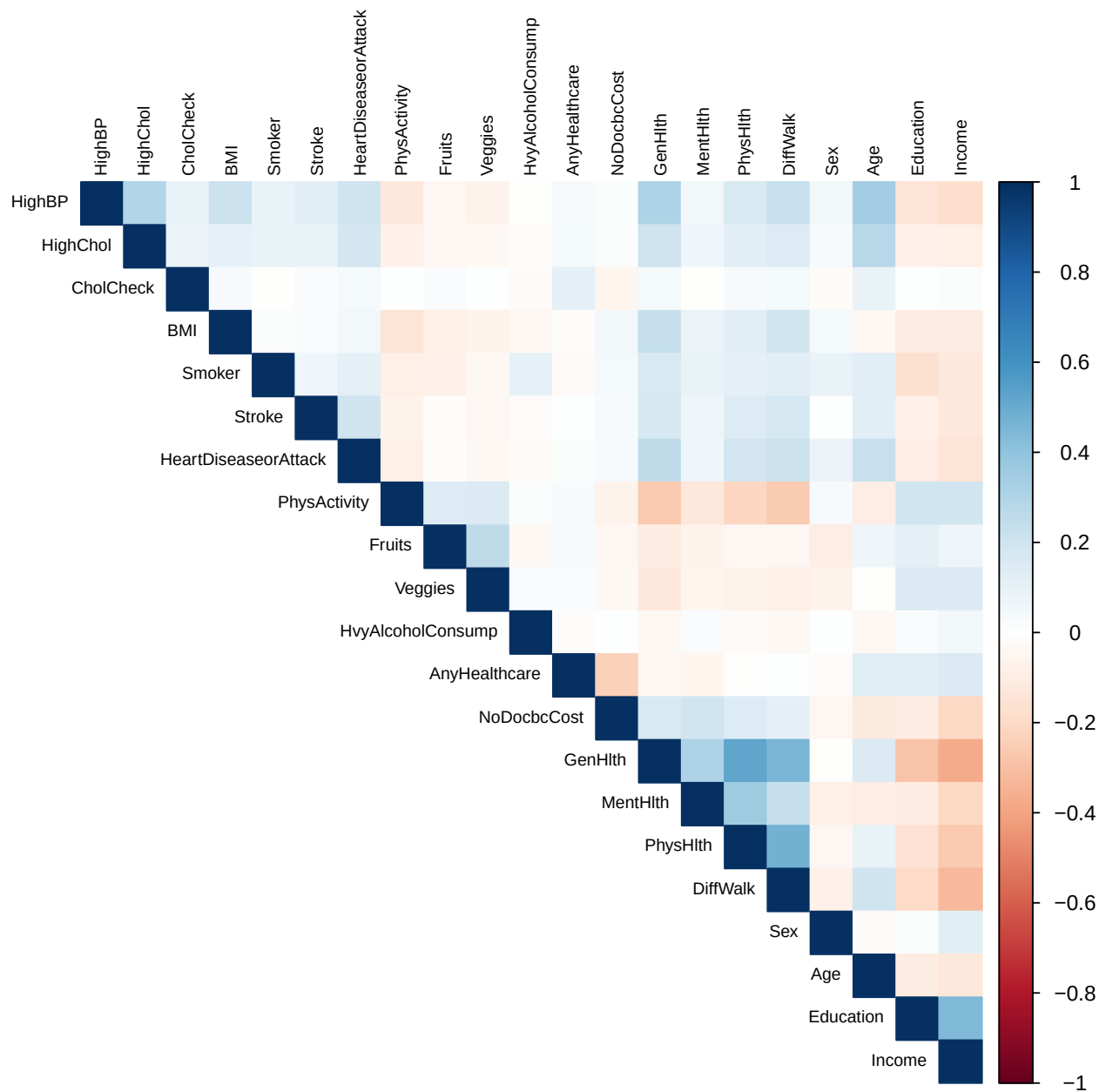
Exploratory Data Analysis (EDA)

Let's look at the distribution of the target variable.



Notice that there is significant class imbalance, only 14.5 percent of interviewed people had diabetes. We will look at up/downweighting to deal with the class imbalance.

We can also examine the correlation between numeric predictors to identify potential multicollinearity.



```
##
## --- Weak Correlations (0.2-0.4) ---
## GenHlth - Income: -0.37
## MentHlth - PhysHlth: 0.35
## Age - HighBP: 0.34
## DiffWalk - Income: -0.32
## GenHlth - MentHlth: 0.30
## GenHlth - HighBP: 0.30
## HighBP - HighChol: 0.30
## Education - GenHlth: -0.28
## Age - HighChol: 0.27
## Income - PhysHlth: -0.27
## GenHlth - PhysActivity: -0.27
## GenHlth - HeartDiseaseorAttack: 0.26
```

```

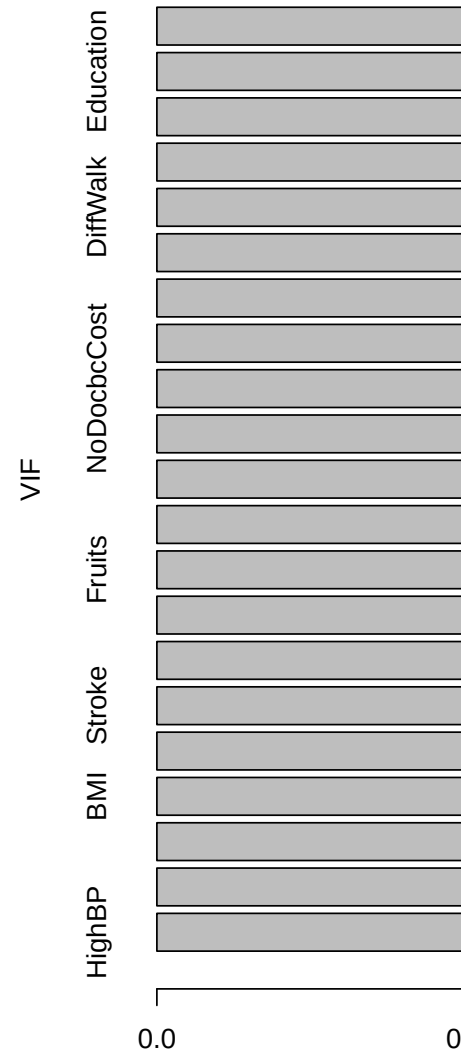
## Fruits - Veggies: 0.25
## DiffWalk - PhysActivity: -0.25
## BMI - GenHlth: 0.24
## DiffWalk - MentHlth: 0.23
## AnyHealthcare - NoDocbcCost: -0.23
## DiffWalk - HighBP: 0.22
## Age - HeartDiseaseorAttack: 0.22
## PhysActivity - PhysHlth: -0.22
## BMI - HighBP: 0.21
## DiffWalk - HeartDiseaseorAttack: 0.21
## Income - MentHlth: -0.21
## HeartDiseaseorAttack - HighBP: 0.21
## GenHlth - HighChol: 0.21
## Age - DiffWalk: 0.20
## Income - NoDocbcCost: -0.20
## HeartDiseaseorAttack - Stroke: 0.20

##
## --- Moderate Correlations (0.4-0.6) ---
## GenHlth - PhysHlth: 0.52
## DiffWalk - PhysHlth: 0.48
## DiffWalk - GenHlth: 0.46
## Education - Income: 0.45

##
## --- No Strong Correlations found (0.6-1) ---

```

The strongest correlations are between GenHlth/PhysHlth/DiffWalk, and Education/Income. This makes sense, as people with better general health likely have better physical health and less difficulty walking. And people with higher education are likely to have a higher income.



We also calculate the variance inflation factor (VIF) to better identify multicollinearity. It seems that the VIFs are relatively small, so we do not need to remove any variables.

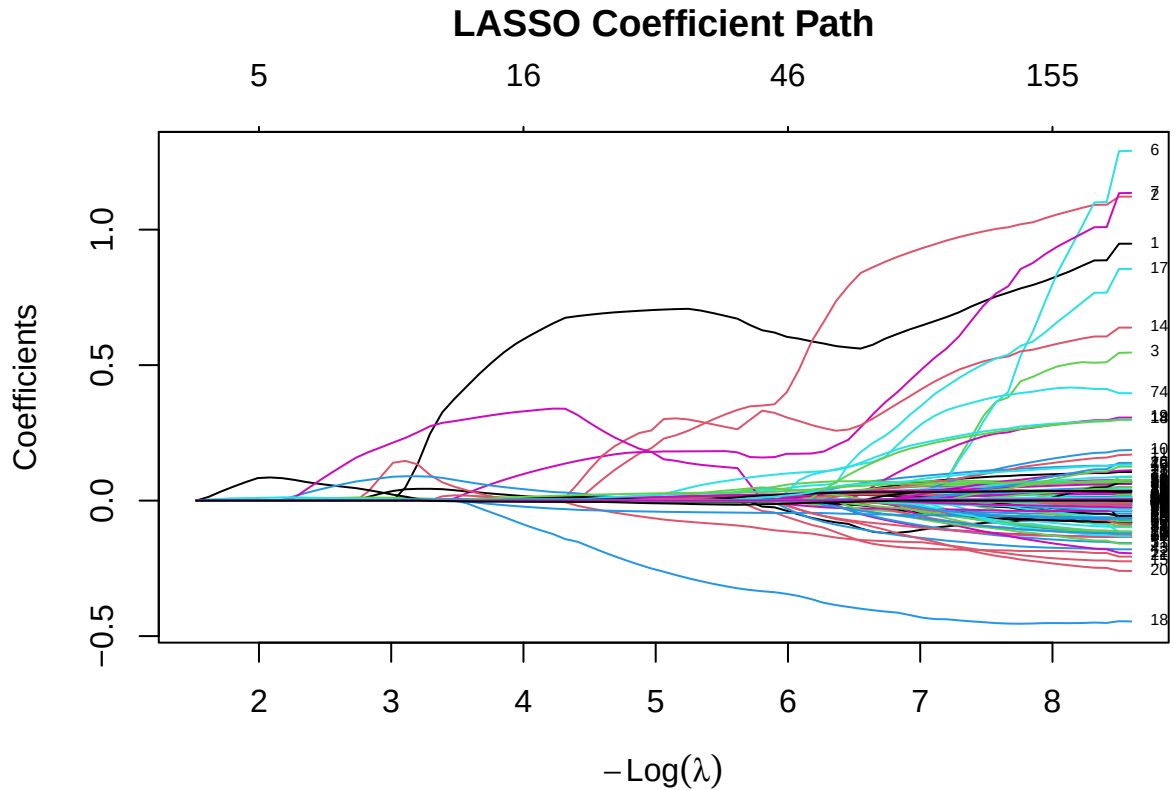
Methodology

We will use K-fold Cross-Validation (5-fold) for model training and evaluation to ensure stability. We will also address class imbalance using upsampling and downsampling techniques. Furthermore, we will use LASSO to select the best variables, including interaction and quadratic terms.

Modeling

We will train three types of models: 1. **Logistic Regression** (Baseline) 2. **Logistic Regression with Class Balancing** (Up/Downsampling) 3. **Logistic Regression with LASSO**

Logistic Regression



##	HighChol	HeartDiseaseorAttack
##	1.063607e+00	9.522356e-01
##	Stroke	HighBP
##	8.936003e-01	8.392018e-01
##	DiffWalk	GenHlth
##	6.808794e-01	5.846811e-01
##	CholCheck	CholCheck:AnyHealthcare
##	5.011143e-01	4.163923e-01
##	HeartDiseaseorAttack:DiffWalk	Age
##	2.869022e-01	2.850407e-01
##	HvyAlcoholConsump:Sex	Smoker:HvyAlcoholConsump
##	2.827400e-01	1.618810e-01
##	Stroke:HeartDiseaseorAttack	BMI
##	1.445333e-01	1.252256e-01
##	DiffWalk:Sex	HighBP:HighChol
##	1.212212e-01	9.876615e-02
##	Fruits:AnyHealthcare	HeartDiseaseorAttack:HvyAlcoholConsump
##	9.679688e-02	9.512012e-02
##	AnyHealthcare:Sex	HeartDiseaseorAttack:Fruits
##	8.764605e-02	7.699006e-02
##	HighBP:NoDocbcCost	HighChol:NoDocbcCost
##	7.045959e-02	6.809786e-02
##	Sex:Age	Smoker:Fruits
##	6.795345e-02	6.759069e-02
##	HighChol:Stroke	HighBP:Fruits
##	6.631324e-02	6.371205e-02

##	PhysActivity:GenHlth	PhysActivity:Sex
##	5.639546e-02	4.282470e-02
##	Smoker:NoDocbcCost	GenHlth:Income
##	3.723793e-02	3.559855e-02
##	GenHlth:Education	Fruits:NoDocbcCost
##	3.557394e-02	3.519885e-02
##	NoDocbcCost:Sex	Stroke:Education
##	3.475156e-02	3.428601e-02
##	HighChol:Smoker	HeartDiseaseorAttack:NoDocbcCost
##	3.188192e-02	3.158079e-02
##	CholCheck:Fruits	HighBP:PhysActivity
##	3.041964e-02	3.036732e-02
##	Veggies:Sex	Smoker:Education
##	3.003471e-02	2.827099e-02
##	Veggies:HvyAlcoholConsump	HighBP:Education
##	2.353132e-02	2.332417e-02
##	CholCheck:BMI	Sex:Income
##	1.722026e-02	1.100005e-02
##	Veggies:Age	NoDocbcCost:Age
##	1.015911e-02	9.043611e-03
##	PhysHlth	NoDocbcCost:Income
##	8.597657e-03	7.591079e-03
##	HvyAlcoholConsump:MentHlth	Stroke:Income
##	6.059339e-03	5.854896e-03
##	HighChol:MentHlth	PhysHlth:Sex
##	5.758663e-03	5.622652e-03
##	AnyHealthcare:PhysHlth	Age:Income
##	4.868278e-03	4.424999e-03
##	HighChol:Veggies	NoDocbcCost:PhysHlth
##	3.649410e-03	3.239650e-03
##	Stroke:MentHlth	Smoker:PhysHlth
##	3.152727e-03	3.121262e-03
##	PhysHlth:DiffWalk	BMI:Age
##	3.096053e-03	2.917074e-03
##	Stroke:PhysHlth	Sex:Education
##	2.757245e-03	2.425944e-03
##	PhysActivity:DiffWalk	HighBP:PhysHlth
##	2.354111e-03	2.082142e-03
##	CholCheck:Income	HeartDiseaseorAttack:MentHlth
##	1.880167e-03	1.803163e-03
##	HighBP:MentHlth	Smoker:Age
##	1.695063e-03	1.685271e-03
##	Veggies:MentHlth	NoDocbcCost:Education
##	1.682983e-03	1.273169e-03
##	BMI:PhysActivity	HighChol:PhysHlth
##	1.197659e-03	1.118862e-03
##	Fruits:MentHlth	HighChol:DiffWalk
##	9.116008e-04	8.297990e-04
##	HeartDiseaseorAttack:PhysHlth	BMI:Smoker
##	5.232971e-04	5.215975e-04
##	AnyHealthcare:GenHlth	I (MentHlth^2)
##	4.714979e-04	3.556358e-04
##	MentHlth:Age	Smoker:Income
##	2.462792e-04	8.033045e-05

##	PhysActivity:MentHlth	BMI:PhysHlth
##	5.852194e-05	3.049515e-05
##	MentHlth:PhysHlth	BMI:Income
##	2.180593e-05	5.235609e-06
##	Smoker	PhysActivity
##	0.000000e+00	0.000000e+00
##	Fruits	Veggies
##	0.000000e+00	0.000000e+00
##	HvyAlcoholConsump	AnyHealthcare
##	0.000000e+00	0.000000e+00
##	NoDocbcCost	MentHlth
##	0.000000e+00	0.000000e+00
##	Sex	I(GenHlth^2)
##	0.000000e+00	0.000000e+00
##	HighBP:CholCheck	HighBP:AnyHealthcare
##	0.000000e+00	0.000000e+00
##	HighBP:DiffWalk	HighBP:Age
##	0.000000e+00	0.000000e+00
##	HighBP:Income	HighChol:CholCheck
##	0.000000e+00	0.000000e+00
##	HighChol:BMI	HighChol:Fruits
##	0.000000e+00	0.000000e+00
##	HighChol:AnyHealthcare	HighChol:GenHlth
##	0.000000e+00	0.000000e+00
##	HighChol:Education	CholCheck:Smoker
##	0.000000e+00	0.000000e+00
##	CholCheck:Stroke	CholCheck:HeartDiseaseorAttack
##	0.000000e+00	0.000000e+00
##	CholCheck:PhysActivity	CholCheck:Veggies
##	0.000000e+00	0.000000e+00
##	CholCheck:HvyAlcoholConsump	CholCheck:NoDocbcCost
##	0.000000e+00	0.000000e+00
##	CholCheck:GenHlth	CholCheck:PhysHlth
##	0.000000e+00	0.000000e+00
##	CholCheck:DiffWalk	CholCheck:Sex
##	0.000000e+00	0.000000e+00
##	CholCheck:Age	CholCheck:Education
##	0.000000e+00	0.000000e+00
##	BMI:HeartDiseaseorAttack	BMI:Fruits
##	0.000000e+00	0.000000e+00
##	BMI:Veggies	BMI:AnyHealthcare
##	0.000000e+00	0.000000e+00
##	BMI:NoDocbcCost	BMI:GenHlth
##	0.000000e+00	0.000000e+00
##	BMI:MentHlth	BMI:Education
##	0.000000e+00	0.000000e+00
##	Smoker:PhysActivity	Smoker:Veggies
##	0.000000e+00	0.000000e+00
##	Smoker:MentHlth	Smoker:DiffWalk
##	0.000000e+00	0.000000e+00
##	Stroke:Fruits	Stroke:AnyHealthcare
##	0.000000e+00	0.000000e+00
##	Stroke:DiffWalk	HeartDiseaseorAttack:Veggies
##	0.000000e+00	0.000000e+00

##	HeartDiseaseorAttack:AnyHealthcare	HeartDiseaseorAttack:Sex
##	0.000000e+00	0.000000e+00
##	HeartDiseaseorAttack:Education	HeartDiseaseorAttack:Income
##	0.000000e+00	0.000000e+00
##	PhysActivity:Veggies	PhysActivity:AnyHealthcare
##	0.000000e+00	0.000000e+00
##	PhysActivity:Income	Fruits:GenHlth
##	0.000000e+00	0.000000e+00
##	Fruits:Income	Veggies:AnyHealthcare
##	0.000000e+00	0.000000e+00
##	Veggies:NoDocbcCost	Veggies:GenHlth
##	0.000000e+00	0.000000e+00
##	HvyAlcoholConsump:NoDocbcCost	HvyAlcoholConsump:Education
##	0.000000e+00	0.000000e+00
##	HvyAlcoholConsump:Income	AnyHealthcare:DiffWalk
##	0.000000e+00	0.000000e+00
##	AnyHealthcare:Age	AnyHealthcare:Income
##	0.000000e+00	0.000000e+00
##	MentHlth:DiffWalk	MentHlth:Income
##	0.000000e+00	0.000000e+00
##	PhysHlth:Age	PhysHlth:Education
##	0.000000e+00	0.000000e+00
##	DiffWalk:Education	Age:Education
##	0.000000e+00	0.000000e+00
##	Education:Income	Fruits:Education
##	0.000000e+00	-1.612914e-04
##	PhysActivity:Fruits	PhysHlth:Income
##	-3.132056e-04	-6.195799e-04
##	MentHlth:Education	GenHlth:MentHlth
##	-6.292653e-04	-6.462181e-04
##	NoDocbcCost:MentHlth	HighBP: BMI
##	-6.978780e-04	-7.310351e-04
##	Fruits:PhysHlth	BMI:Sex
##	-9.573596e-04	-1.306445e-03
##	BMI:HvyAlcoholConsump	I (BMI^2)
##	-1.351657e-03	-1.418989e-03
##	Veggies:PhysHlth	BMI:DiffWalk
##	-1.430316e-03	-1.514151e-03
##	PhysActivity:PhysHlth	Stroke:Sex
##	-1.595990e-03	-1.599955e-03
##	HvyAlcoholConsump:GenHlth	Veggies:Education
##	-1.647279e-03	-2.134155e-03
##	Veggies:Income	I (Income^2)
##	-2.315514e-03	-2.442073e-03
##	HvyAlcoholConsump:PhysHlth	HighChol:Income
##	-2.490131e-03	-3.016026e-03
##	MentHlth:Sex	DiffWalk:Income
##	-3.222420e-03	-3.474614e-03
##	AnyHealthcare:MentHlth	HighChol:PhysActivity
##	-3.571794e-03	-5.325983e-03
##	PhysActivity:Age	Fruits:Age
##	-5.368812e-03	-5.508026e-03
##	BMI:Stroke	GenHlth:PhysHlth
##	-5.626453e-03	-5.907470e-03

##	HighBP:Smoker	I(Age^2)
##	-7.957385e-03	-1.102659e-02
##	Fruits:DiffWalk	CholCheck:MentHlth
##	-1.190184e-02	-1.275734e-02
##	Fruits:HvyAlcoholConsump	Smoker:Sex
##	-1.281496e-02	-1.520568e-02
##	HeartDiseaseorAttack:Age	DiffWalk:Age
##	-1.886934e-02	-2.067249e-02
##	Veggies:DiffWalk	NoDocbcCost:GenHlth
##	-2.155956e-02	-2.246901e-02
##	GenHlth:Age	HighChol:HvyAlcoholConsump
##	-2.257268e-02	-2.320538e-02
##	PhysActivity:HvyAlcoholConsump	Smoker:HeartDiseaseorAttack
##	-2.947349e-02	-3.063822e-02
##	PhysActivity:Education	Stroke:Age
##	-3.857962e-02	-4.159770e-02
##	PhysActivity:NoDocbcCost	HighBP:HvyAlcoholConsump
##	-4.660175e-02	-4.779394e-02
##	Smoker:AnyHealthcare	Stroke:GenHlth
##	-5.332372e-02	-5.406492e-02
##	Stroke:NoDocbcCost	HighChol:Age
##	-5.651359e-02	-5.933192e-02
##	HvyAlcoholConsump:Age	HvyAlcoholConsump:DiffWalk
##	-6.044812e-02	-6.317834e-02
##	AnyHealthcare:Education	Education
##	-6.586458e-02	-7.072129e-02
##	HighBP:Veggies	Smoker:GenHlth
##	-7.270445e-02	-7.405518e-02
##	HighBP:GenHlth	HighBP:Stroke
##	-7.445494e-02	-8.632751e-02
##	Stroke:Veggies	Fruits:Sex
##	-1.002182e-01	-1.029939e-01
##	HeartDiseaseorAttack:PhysActivity	Smoker:Stroke
##	-1.068394e-01	-1.092890e-01
##	GenHlth:DiffWalk	GenHlth:Sex
##	-1.117738e-01	-1.149030e-01
##	HeartDiseaseorAttack:GenHlth	AnyHealthcare:NoDocbcCost
##	-1.156798e-01	-1.162605e-01
##	HighBP:HeartDiseaseorAttack	HighChol:Sex
##	-1.301247e-01	-1.308358e-01
##	Stroke:PhysActivity	HighChol:HeartDiseaseorAttack
##	-1.348529e-01	-1.461160e-01
##	Stroke:HvyAlcoholConsump	HighBP:Sex
##	-1.637164e-01	-1.750148e-01
##	Income	Fruits:Veggies
##	-1.869691e-01	-2.153779e-01
##	NoDocbcCost:DiffWalk	HvyAlcoholConsump:AnyHealthcare
##	-2.369516e-01	-4.520690e-01

Select 1SE sparser model.

##	HighChol	HighBP
##	0.9254825249	0.6410146608
##	HeartDiseaseorAttack	GenHlth

##	0.4696153201	0.4021505243
##	DiffWalk	CholCheck:AnyHealthcare
##	0.3389563766	0.2700373779
##	HeartDiseaseorAttack:DiffWalk	HvyAlcoholConsump:Sex
##	0.2086044159	0.1983377768
##	Age	BMI
##	0.1390861526	0.0893871757
##	DiffWalk:Sex	HighChol:NoDocbcCost
##	0.0769275099	0.0704413492
##	HighBP:HighChol	Sex:Age
##	0.0665547447	0.0628802738
##	Stroke:HeartDiseaseorAttack	Smoker:Fruits
##	0.0600867326	0.0547214307
##	HighBP:Fruits	HeartDiseaseorAttack:Fruits
##	0.0515469037	0.0408485009
##	PhysActivity:GenHlth	GenHlth:Education
##	0.0368602489	0.0359447375
##	HeartDiseaseorAttack:NoDocbcCost	GenHlth:Income
##	0.0318374715	0.0316571430
##	CholCheck:BMI	Stroke:Education
##	0.0293533447	0.0279805268
##	Smoker:HvyAlcoholConsump	HighBP:Education
##	0.0276812017	0.0267498978
##	HighBP:NoDocbcCost	CholCheck:Age
##	0.0259038458	0.0231142818
##	HeartDiseaseorAttack:HvyAlcoholConsump	HighBP:PhysActivity
##	0.0208540688	0.0153054502
##	Smoker:Education	HighChol:Smoker
##	0.0105217538	0.0089403384
##	PhysActivity:Sex	HighChol:MentHlth
##	0.0063965893	0.0051413144
##	Stroke:MentHlth	BMI:Age
##	0.0038301568	0.0038085827
##	Age:Income	PhysHlth
##	0.0032548028	0.0028840011
##	PhysHlth:Sex	Sex:Income
##	0.0022238667	0.0020966089
##	Stroke:Income	HvyAlcoholConsump:MentHlth
##	0.0012873340	0.0012065258
##	HighChol:PhysHlth	Smoker:PhysHlth
##	0.0010634599	0.0009884163
##	HeartDiseaseorAttack:MentHlth	AnyHealthcare:PhysHlth
##	0.0008593301	0.0007214732
##	PhysHlth:DiffWalk	Fruits:AnyHealthcare
##	0.0004061470	0.0002172722
##	CholCheck	Smoker
##	0.0000000000	0.0000000000
##	Stroke	PhysActivity
##	0.0000000000	0.0000000000
##	Fruits	Veggies
##	0.0000000000	0.0000000000
##	HvyAlcoholConsump	AnyHealthcare
##	0.0000000000	0.0000000000
##	NoDocbcCost	MentHlth

##	0.0000000000	0.0000000000
##	Sex	I(MentHlth^2)
##	0.0000000000	0.0000000000
##	I(GenHlth^2)	HighBP:CholCheck
##	0.0000000000	0.0000000000
##	HighBP:BMI	HighBP:Smoker
##	0.0000000000	0.0000000000
##	HighBP:Stroke	HighBP:Veggies
##	0.0000000000	0.0000000000
##	HighBP:AnyHealthcare	HighBP:MentHlth
##	0.0000000000	0.0000000000
##	HighBP:PhysHlth	HighBP:DiffWalk
##	0.0000000000	0.0000000000
##	HighBP:Age	HighBP:Income
##	0.0000000000	0.0000000000
##	HighChol:CholCheck	HighChol:BMI
##	0.0000000000	0.0000000000
##	HighChol:Stroke	HighChol:PhysActivity
##	0.0000000000	0.0000000000
##	HighChol:Fruits	HighChol:Veggies
##	0.0000000000	0.0000000000
##	HighChol:HvyAlcoholConsump	HighChol:AnyHealthcare
##	0.0000000000	0.0000000000
##	HighChol:GenHlth	HighChol:DiffWalk
##	0.0000000000	0.0000000000
##	HighChol:Education	HighChol:Income
##	0.0000000000	0.0000000000
##	CholCheck:Smoker	CholCheck:Stroke
##	0.0000000000	0.0000000000
##	CholCheck:HeartDiseaseorAttack	CholCheck:PhysActivity
##	0.0000000000	0.0000000000
##	CholCheck:Fruits	CholCheck:Veggies
##	0.0000000000	0.0000000000
##	CholCheck:HvyAlcoholConsump	CholCheck:NoDocbcCost
##	0.0000000000	0.0000000000
##	CholCheck:GenHlth	CholCheck:PhysHlth
##	0.0000000000	0.0000000000
##	CholCheck:DiffWalk	CholCheck:Sex
##	0.0000000000	0.0000000000
##	CholCheck:Education	CholCheck:Income
##	0.0000000000	0.0000000000
##	BMI:Smoker	BMI:Stroke
##	0.0000000000	0.0000000000
##	BMI:HeartDiseaseorAttack	BMI:PhysActivity
##	0.0000000000	0.0000000000
##	BMI:Fruits	BMI:Veggies
##	0.0000000000	0.0000000000
##	BMI:HvyAlcoholConsump	BMI:AnyHealthcare
##	0.0000000000	0.0000000000
##	BMI:NoDocbcCost	BMI:GenHlth
##	0.0000000000	0.0000000000
##	BMI:MentHlth	BMI:PhysHlth
##	0.0000000000	0.0000000000
##	BMI:DiffWalk	BMI:Sex

##	0.0000000000	0.0000000000
##	BMI:Education	BMI:Income
##	0.0000000000	0.0000000000
##	Smoker:HeartDiseaseorAttack	Smoker:PhysActivity
##	0.0000000000	0.0000000000
##	Smoker:Veggies	Smoker:AnyHealthcare
##	0.0000000000	0.0000000000
##	Smoker:NoDocbcCost	Smoker:MentHlth
##	0.0000000000	0.0000000000
##	Smoker:DiffWalk	Smoker:Sex
##	0.0000000000	0.0000000000
##	Smoker:Age	Smoker:Income
##	0.0000000000	0.0000000000
##	Stroke:Fruits	Stroke:AnyHealthcare
##	0.0000000000	0.0000000000
##	Stroke:NoDocbcCost	Stroke:GenHlth
##	0.0000000000	0.0000000000
##	Stroke:PhysHlth	Stroke:DiffWalk
##	0.0000000000	0.0000000000
##	Stroke:Sex	Stroke:Age
##	0.0000000000	0.0000000000
##	HeartDiseaseorAttack:Veggies	HeartDiseaseorAttack:AnyHealthcare
##	0.0000000000	0.0000000000
##	HeartDiseaseorAttack:PhysHlth	HeartDiseaseorAttack:Sex
##	0.0000000000	0.0000000000
##	HeartDiseaseorAttack:Age	HeartDiseaseorAttack:Education
##	0.0000000000	0.0000000000
##	HeartDiseaseorAttack:Income	PhysActivity:Fruits
##	0.0000000000	0.0000000000
##	PhysActivity:Veggies	PhysActivity:AnyHealthcare
##	0.0000000000	0.0000000000
##	PhysActivity:NoDocbcCost	PhysActivity:MentHlth
##	0.0000000000	0.0000000000
##	PhysActivity:PhysHlth	PhysActivity:DiffWalk
##	0.0000000000	0.0000000000
##	PhysActivity:Age	PhysActivity:Income
##	0.0000000000	0.0000000000
##	Fruits:HvyAlcoholConsump	Fruits:NoDocbcCost
##	0.0000000000	0.0000000000
##	Fruits:GenHlth	Fruits:MentHlth
##	0.0000000000	0.0000000000
##	Fruits:PhysHlth	Fruits:DiffWalk
##	0.0000000000	0.0000000000
##	Fruits:Age	Fruits:Education
##	0.0000000000	0.0000000000
##	Fruits:Income	Veggies:HvyAlcoholConsump
##	0.0000000000	0.0000000000
##	Veggies:AnyHealthcare	Veggies:NoDocbcCost
##	0.0000000000	0.0000000000
##	Veggies:GenHlth	Veggies:MentHlth
##	0.0000000000	0.0000000000
##	Veggies:PhysHlth	Veggies:DiffWalk
##	0.0000000000	0.0000000000
##	Veggies:Sex	Veggies:Age

##	0.0000000000	0.0000000000
##	Veggies:Education	Veggies:Income
##	0.0000000000	0.0000000000
##	HvyAlcoholConsump:NoDocbcCost	HvyAlcoholConsump:GenHlth
##	0.0000000000	0.0000000000
##	HvyAlcoholConsump:PhysHlth	HvyAlcoholConsump:Education
##	0.0000000000	0.0000000000
##	HvyAlcoholConsump:Income	AnyHealthcare:GenHlth
##	0.0000000000	0.0000000000
##	AnyHealthcare:DiffWalk	AnyHealthcare:Sex
##	0.0000000000	0.0000000000
##	AnyHealthcare:Age	AnyHealthcare:Income
##	0.0000000000	0.0000000000
##	NoDocbcCost:GenHlth	NoDocbcCost:MentHlth
##	0.0000000000	0.0000000000
##	NoDocbcCost:PhysHlth	NoDocbcCost:Sex
##	0.0000000000	0.0000000000
##	NoDocbcCost:Age	NoDocbcCost:Education
##	0.0000000000	0.0000000000
##	NoDocbcCost:Income	GenHlth:MentHlth
##	0.0000000000	0.0000000000
##	MentHlth:PhysHlth	MentHlth:DiffWalk
##	0.0000000000	0.0000000000
##	MentHlth:Age	MentHlth:Education
##	0.0000000000	0.0000000000
##	MentHlth:Income	PhysHlth:Age
##	0.0000000000	0.0000000000
##	PhysHlth:Education	PhysHlth:Income
##	0.0000000000	0.0000000000
##	DiffWalk:Education	DiffWalk:Income
##	0.0000000000	0.0000000000
##	Sex:Education	Age:Education
##	0.0000000000	0.0000000000
##	Education:Income	I (BMI^2)
##	0.0000000000	-0.0011845539
##	Stroke:Veggies	I (Income^2)
##	-0.0012113504	-0.0013854277
##	CholCheck:MentHlth	AnyHealthcare:MentHlth
##	-0.0020870518	-0.0024790622
##	GenHlth:PhysHlth	MentHlth:Sex
##	-0.0031076650	-0.0033843997
##	PhysActivity:HvyAlcoholConsump	DiffWalk:Age
##	-0.0054413176	-0.0058680843
##	I (Age^2)	GenHlth:Age
##	-0.0071344982	-0.0117315470
##	HeartDiseaseorAttack:PhysActivity	AnyHealthcare:NoDocbcCost
##	-0.0157140071	-0.0160636455
##	Smoker:Stroke	HighBP:HvyAlcoholConsump
##	-0.0182012814	-0.0228946332
##	AnyHealthcare:Education	HighBP:GenHlth
##	-0.0240989343	-0.0265234152
##	HvyAlcoholConsump:DiffWalk	PhysActivity:Education
##	-0.0287616884	-0.0307843447
##	Fruits:Sex	Stroke:PhysActivity

```
##                -0.0352604840                -0.0386518307
##                Smoker:GenHlth                HighChol:Age
##                -0.0427398031                -0.0441998500
##                Stroke:HvyAlcoholConsump                HvyAlcoholConsump:Age
##                -0.0500103434                -0.0533306156
##                HeartDiseaseorAttack:GenHlth                GenHlth:Sex
##                -0.0556715376                -0.0627117810
##                GenHlth:DiffWalk                HighChol:HeartDiseaseorAttack
##                -0.0660947082                -0.0890972193
##                HighBP:HeartDiseaseorAttack                HighChol:Sex
##                -0.0930875510                -0.0985742304
##                Education                HighBP:Sex
##                -0.1098785461                -0.1361039407
##                NoDocbcCost:DiffWalk                Fruits:Veggies
##                -0.1376834884                -0.1533827954
##                Income                HvyAlcoholConsump:AnyHealthcare
##                -0.1757868311                -0.4281189896
```

We only take coefficients with absolute values greater than 0.1. This is nonstandard, but we do this to identify a very sparse model.

```
##                HighChol                HighBP
##                0.9254825                0.6410147
##                HeartDiseaseorAttack                GenHlth
##                0.4696153                0.4021505
##                DiffWalk                CholCheck:AnyHealthcare
##                0.3389564                0.2700374
##                HeartDiseaseorAttack:DiffWalk                HvyAlcoholConsump:Sex
##                0.2086044                0.1983378
##                Age                Education
##                0.1390862                -0.1098785
##                HighBP:Sex                NoDocbcCost:DiffWalk
##                -0.1361039                -0.1376835
##                Fruits:Veggies                Income
##                -0.1533828                -0.1757868
##                HvyAlcoholConsump:AnyHealthcare
##                -0.4281190
```

Finally, we refit the model with the selected variables.

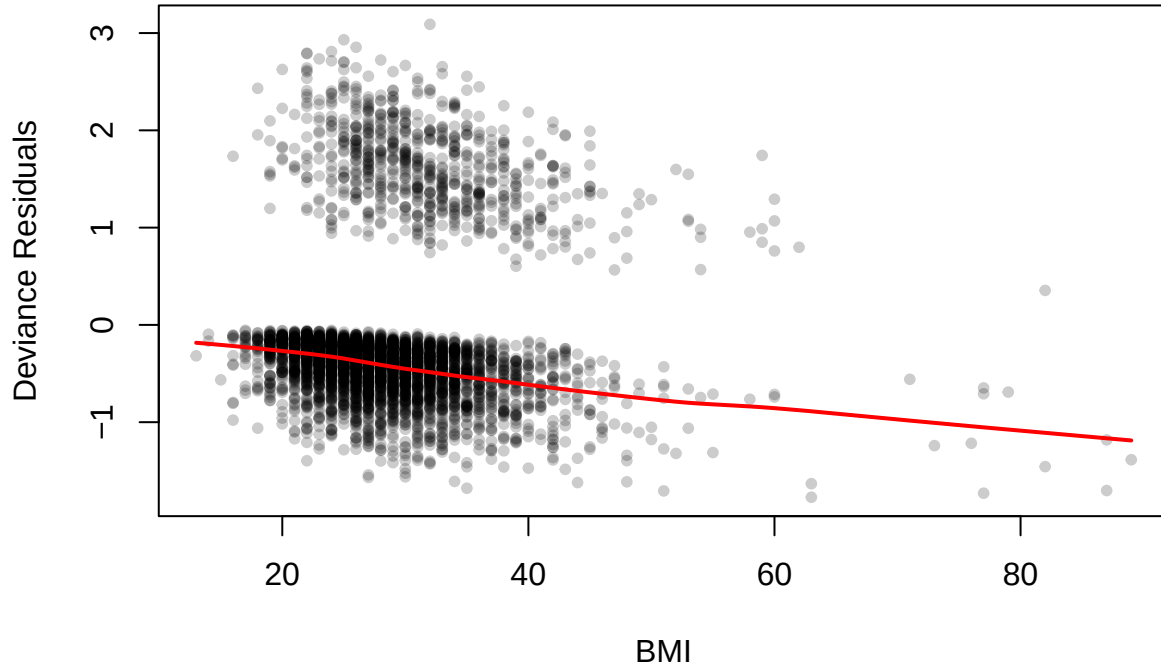
Residual Analysis

Let's examine the residuals of the logistic regression model to check for non-linearities.

Check BMI, MentHlth, GenHlth, PhysHlth, Age, Education, Income

```
## Gut Check: Deviance residuals should be roughly symmetric around 0
## Mean Residual: -0.1489146
```

Deviance Residuals vs BMI



Results and Evaluation

We evaluate the models on the holdout test set using ROC-AUC, Sensitivity (Recall), and Specificity.

Table 1: Model Performance Comparison

Model	AUC	Recall	F1
GLM (Standard)	0.822	0.161	0.248
GLM (Downsample)	0.822	0.771	0.445
GLM (Upsample)	0.822	0.771	0.445
LASSO (Downsample)	0.829	0.789	0.443
LASSO (1SE)	0.807	0.753	0.429

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    No   Yes
##           No 42715 5934
##           Yes  951 1135
##
##           Accuracy : 0.8643
##           95% CI : (0.8613, 0.8673)
##           No Information Rate : 0.8607
##           P-Value [Acc > NIR] : 0.009147
##
##           Kappa : 0.197
##
```



```

## McNemar's Test P-Value : < 2.2e-16
##
##          Sensitivity : 0.16056
##          Specificity : 0.97822
##          Pos Pred Value : 0.54410
##          Neg Pred Value : 0.87802
##          Prevalence : 0.13933
##          Detection Rate : 0.02237
##          Detection Prevalence : 0.04112
##          Balanced Accuracy : 0.56939
##
##          'Positive' Class : Yes
##

## Confusion Matrix and Statistics
##
##          Reference
## Prediction    No    Yes
##          No  31686  1617
##          Yes 11980  5452
##
##          Accuracy : 0.732
##          95% CI : (0.7281, 0.7359)
##          No Information Rate : 0.8607
##          P-Value [Acc > NIR] : 1
##
##          Kappa : 0.3078
##
## McNemar's Test P-Value : <2e-16
##
##          Sensitivity : 0.7713
##          Specificity : 0.7256
##          Pos Pred Value : 0.3128
##          Neg Pred Value : 0.9514
##          Prevalence : 0.1393
##          Detection Rate : 0.1075
##          Detection Prevalence : 0.3436
##          Balanced Accuracy : 0.7484
##
##          'Positive' Class : Yes
##

## Confusion Matrix and Statistics
##
##          Reference
## Prediction    No    Yes
##          No  31707  1616
##          Yes 11959  5453
##
##          Accuracy : 0.7324
##          95% CI : (0.7286, 0.7363)
##          No Information Rate : 0.8607
##          P-Value [Acc > NIR] : 1
##

```

```

##                Kappa : 0.3084
##
## Mcnemar's Test P-Value : <2e-16
##
##          Sensitivity : 0.7714
##          Specificity : 0.7261
##          Pos Pred Value : 0.3132
##          Neg Pred Value : 0.9515
##          Prevalence : 0.1393
##          Detection Rate : 0.1075
##          Detection Prevalence : 0.3432
##          Balanced Accuracy : 0.7488
##
##          'Positive' Class : Yes
##

## Confusion Matrix and Statistics
##
##          Reference
## Prediction    No    Yes
##          No  31157  1494
##          Yes 12509  5575
##
##          Accuracy : 0.724
##          95% CI : (0.7201, 0.7279)
##          No Information Rate : 0.8607
##          P-Value [Acc > NIR] : 1
##
##          Kappa : 0.3038
##
## Mcnemar's Test P-Value : <2e-16
##
##          Sensitivity : 0.7887
##          Specificity : 0.7135
##          Pos Pred Value : 0.3083
##          Neg Pred Value : 0.9542
##          Prevalence : 0.1393
##          Detection Rate : 0.1099
##          Detection Prevalence : 0.3564
##          Balanced Accuracy : 0.7511
##
##          'Positive' Class : Yes
##

## Confusion Matrix and Statistics
##
##          Reference
## Prediction    No    Yes
##          No  31245  1746
##          Yes 12421  5323
##
##          Accuracy : 0.7208
##          95% CI : (0.7168, 0.7247)
##          No Information Rate : 0.8607

```

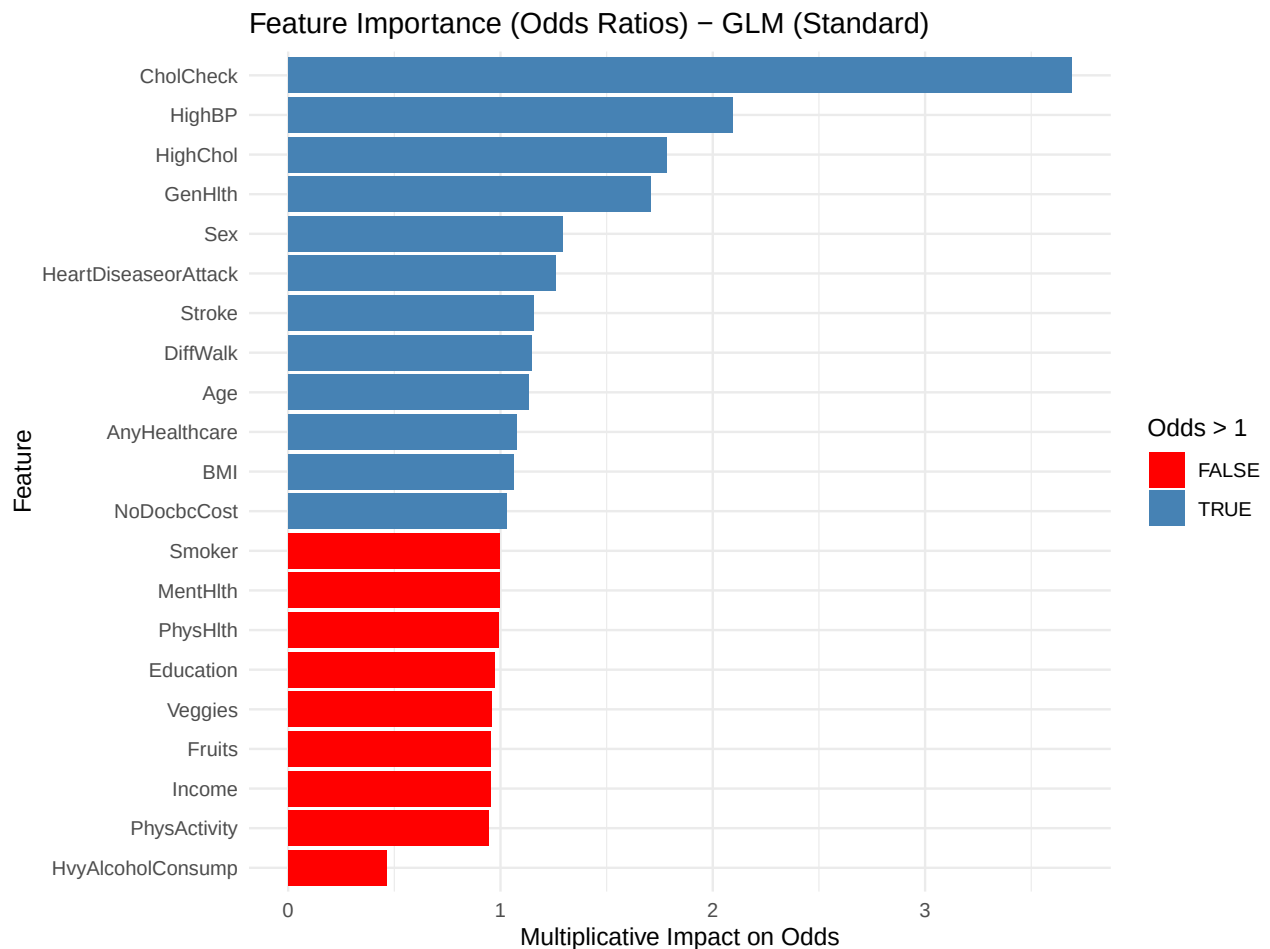
```

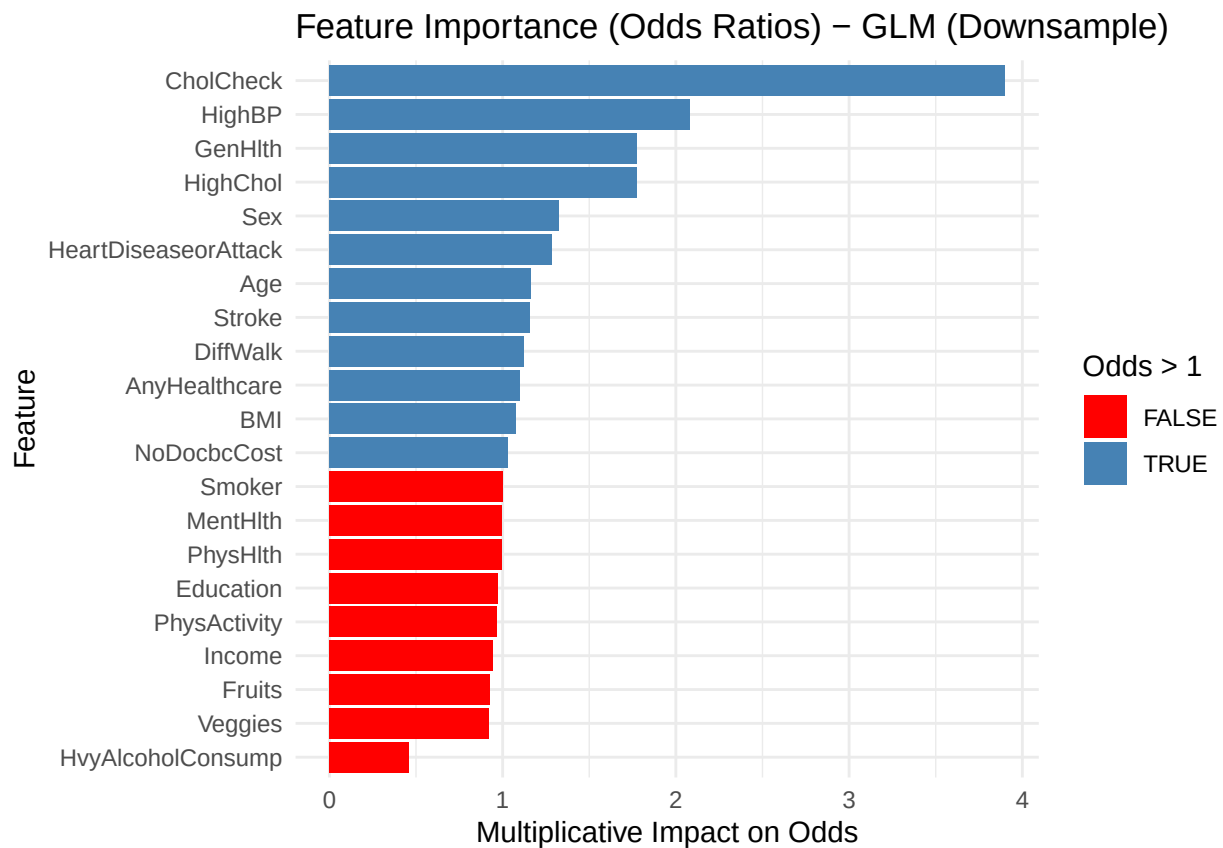
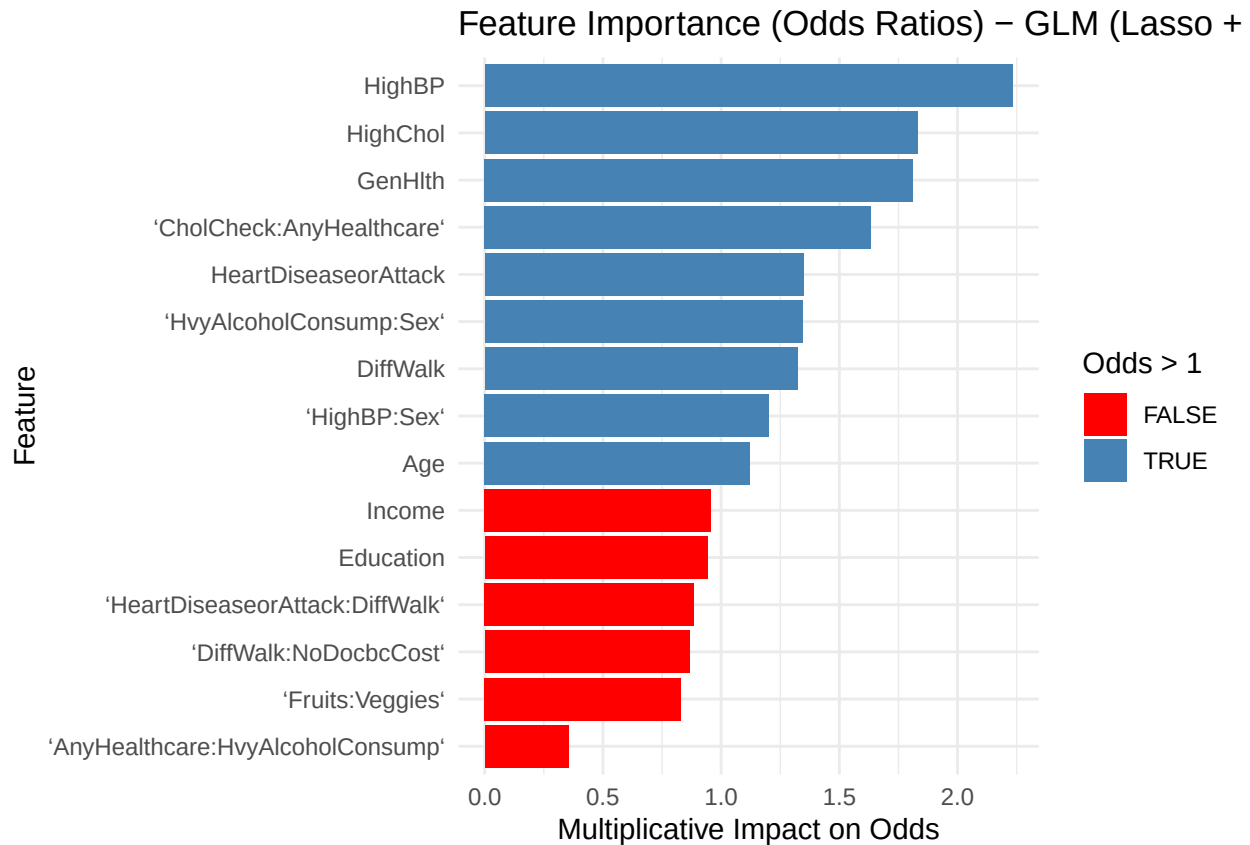
##      P-Value [Acc > NIR] : 1
##
##              Kappa : 0.287
##
##  McNemar's Test P-Value : <2e-16
##
##      Sensitivity : 0.7530
##      Specificity : 0.7155
##      Pos Pred Value : 0.3000
##      Neg Pred Value : 0.9471
##      Prevalence : 0.1393
##      Detection Rate : 0.1049
##      Detection Prevalence : 0.3497
##      Balanced Accuracy : 0.7343
##
##      'Positive' Class : Yes
##

```

Feature Importance

We analyze the multiplicative impact of features on the odds of diabetes using the coefficients from each of the models.





Fairness Analysis

We examine the model's performance across different demographic groups to ensure fairness. We will use the Downsampled GLM for this analysis.

```
##
## ### Fairness Analysis by Sex
##
## #### Group: 0
##           Reference
## Prediction    No   Yes
##           No 18616  858
##           Yes  6121 2828
## Sensitivity: 0.767
## Specificity: 0.753
## Balanced Accuracy: 0.76
##
## #### Group: 1
##           Reference
## Prediction    No   Yes
##           No 13070  759
##           Yes  5859 2624
## Sensitivity: 0.776
## Specificity: 0.69
## Balanced Accuracy: 0.733
##
## ### Fairness Analysis by Age
##
## #### Group: 1
##           Reference
## Prediction    No   Yes
##           No  1132   14
##           Yes    17    1
## Sensitivity: 0.067
## Specificity: 0.985
## Balanced Accuracy: 0.526
##
## #### Group: 2
##           Reference
## Prediction    No   Yes
##           No 1427   18
##           Yes   36   12
## Sensitivity: 0.4
## Specificity: 0.975
## Balanced Accuracy: 0.688
##
## #### Group: 3
##           Reference
## Prediction    No   Yes
##           No 2045   43
##           Yes   98   20
## Sensitivity: 0.317
## Specificity: 0.954
## Balanced Accuracy: 0.636
```

```

##
## ##### Group: 4
##           Reference
## Prediction  No  Yes
##           No 2473  83
##           Yes 166  70
## Sensitivity: 0.458
## Specificity: 0.937
## Balanced Accuracy: 0.697
##
## ##### Group: 5
##           Reference
## Prediction  No  Yes
##           No 2731  96
##           Yes 299 107
## Sensitivity: 0.527
## Specificity: 0.901
## Balanced Accuracy: 0.714
##
## ##### Group: 6
##           Reference
## Prediction  No  Yes
##           No 3050 128
##           Yes 543 221
## Sensitivity: 0.633
## Specificity: 0.849
## Balanced Accuracy: 0.741
##
## ##### Group: 7
##           Reference
## Prediction  No  Yes
##           No 3723 201
##           Yes 870 415
## Sensitivity: 0.674
## Specificity: 0.811
## Balanced Accuracy: 0.742
##
## ##### Group: 8
##           Reference
## Prediction  No  Yes
##           No 3992 228
##           Yes 1309 627
## Sensitivity: 0.733
## Specificity: 0.753
## Balanced Accuracy: 0.743
##
## ##### Group: 9
##           Reference
## Prediction  No  Yes
##           No 3771 256
##           Yes 1773 924
## Sensitivity: 0.783
## Specificity: 0.68
## Balanced Accuracy: 0.732

```

```

##
## ##### Group: 10
##           Reference
## Prediction  No  Yes
##           No 3246 232
##           Yes 1903 1052
## Sensitivity: 0.819
## Specificity: 0.63
## Balanced Accuracy: 0.725
##
## ##### Group: 11
##           Reference
## Prediction  No  Yes
##           No 1885 149
##           Yes 1767 834
## Sensitivity: 0.848
## Specificity: 0.516
## Balanced Accuracy: 0.682
##
## ##### Group: 12
##           Reference
## Prediction  No  Yes
##           No 1131 80
##           Yes 1407 584
## Sensitivity: 0.88
## Specificity: 0.446
## Balanced Accuracy: 0.663
##
## ##### Group: 13
##           Reference
## Prediction  No  Yes
##           No 1080 89
##           Yes 1792 585
## Sensitivity: 0.868
## Specificity: 0.376
## Balanced Accuracy: 0.622
##
## ### Fairness Analysis by Education
##
## ##### Group: 1
##           Reference
## Prediction No Yes
##           No 10 0
##           Yes 14 11
## Sensitivity: 1
## Specificity: 0.417
## Balanced Accuracy: 0.708
##
## ##### Group: 2
##           Reference
## Prediction No Yes
##           No 200 26
##           Yes 350 230
## Sensitivity: 0.898

```

```

## Specificity: 0.364
## Balanced Accuracy: 0.631
##
## ##### Group: 3
##           Reference
## Prediction  No  Yes
##           No  685  46
##           Yes 717 430
## Sensitivity: 0.903
## Specificity: 0.489
## Balanced Accuracy: 0.696
##
## ##### Group: 4
##           Reference
## Prediction  No  Yes
##           No  6432 397
##           Yes 4047 1777
## Sensitivity: 0.817
## Specificity: 0.614
## Balanced Accuracy: 0.716
##
## ##### Group: 5
##           Reference
## Prediction  No  Yes
##           No  8587 494
##           Yes 3368 1632
## Sensitivity: 0.768
## Specificity: 0.718
## Balanced Accuracy: 0.743
##
## ##### Group: 6
##           Reference
## Prediction  No  Yes
##           No  15772 654
##           Yes  3484 1372
## Sensitivity: 0.677
## Specificity: 0.819
## Balanced Accuracy: 0.748
##
## ### Fairness Analysis by Income
##
## ##### Group: 1
##           Reference
## Prediction  No  Yes
##           No  720  51
##           Yes 775 480
## Sensitivity: 0.904
## Specificity: 0.482
## Balanced Accuracy: 0.693
##
## ##### Group: 2
##           Reference
## Prediction  No  Yes
##           No  773  55

```



```

##          Yes 960 541
## Sensitivity: 0.908
## Specificity: 0.446
## Balanced Accuracy: 0.677
##
## ##### Group: 3
##          Reference
## Prediction  No  Yes
##          No 1298  72
##          Yes 1197 621
## Sensitivity: 0.896
## Specificity: 0.52
## Balanced Accuracy: 0.708
##
## ##### Group: 4
##          Reference
## Prediction  No  Yes
##          No 1824 139
##          Yes 1384 697
## Sensitivity: 0.834
## Specificity: 0.569
## Balanced Accuracy: 0.701
##
## ##### Group: 5
##          Reference
## Prediction  No  Yes
##          No 2703 166
##          Yes 1647 769
## Sensitivity: 0.822
## Specificity: 0.621
## Balanced Accuracy: 0.722
##
## ##### Group: 6
##          Reference
## Prediction  No  Yes
##          No 4300 235
##          Yes 1923 812
## Sensitivity: 0.776
## Specificity: 0.691
## Balanced Accuracy: 0.733
##
## ##### Group: 7
##          Reference
## Prediction  No  Yes
##          No 5886 304
##          Yes 1761 715
## Sensitivity: 0.702
## Specificity: 0.77
## Balanced Accuracy: 0.736
##
## ##### Group: 8
##          Reference
## Prediction  No  Yes
##          No 14182 595

```

```

##          Yes  2333   817
## Sensitivity: 0.579
## Specificity: 0.859
## Balanced Accuracy: 0.719

##
## ### Fairness Analysis by Sex
##
## #### Group: 0
##          Reference
## Prediction   No   Yes
##           No 24253 3064
##           Yes  484  622
## Sensitivity: 0.169
## Specificity: 0.98
## Balanced Accuracy: 0.575
##
## #### Group: 1
##          Reference
## Prediction   No   Yes
##           No 18462 2870
##           Yes  467  513
## Sensitivity: 0.152
## Specificity: 0.975
## Balanced Accuracy: 0.563
##
## ### Fairness Analysis by Age
##
## #### Group: 1
##          Reference
## Prediction   No   Yes
##           No 1147   15
##           Yes    2    0
## Sensitivity: 0
## Specificity: 0.998
## Balanced Accuracy: 0.499
##
## #### Group: 2
##          Reference
## Prediction   No   Yes
##           No 1460   27
##           Yes    3    3
## Sensitivity: 0.1
## Specificity: 0.998
## Balanced Accuracy: 0.549
##
## #### Group: 3
##          Reference
## Prediction   No   Yes
##           No 2138   62
##           Yes    5    1
## Sensitivity: 0.016
## Specificity: 0.998
## Balanced Accuracy: 0.507

```

```

##
## ##### Group: 4
##           Reference
## Prediction  No  Yes
##           No 2630 149
##           Yes   9   4
## Sensitivity: 0.026
## Specificity: 0.997
## Balanced Accuracy: 0.511
##
## ##### Group: 5
##           Reference
## Prediction  No  Yes
##           No 3016 192
##           Yes  14  11
## Sensitivity: 0.054
## Specificity: 0.995
## Balanced Accuracy: 0.525
##
## ##### Group: 6
##           Reference
## Prediction  No  Yes
##           No 3562 304
##           Yes  31  45
## Sensitivity: 0.129
## Specificity: 0.991
## Balanced Accuracy: 0.56
##
## ##### Group: 7
##           Reference
## Prediction  No  Yes
##           No 4529 544
##           Yes  64  72
## Sensitivity: 0.117
## Specificity: 0.986
## Balanced Accuracy: 0.551
##
## ##### Group: 8
##           Reference
## Prediction  No  Yes
##           No 5214 718
##           Yes  87 137
## Sensitivity: 0.16
## Specificity: 0.984
## Balanced Accuracy: 0.572
##
## ##### Group: 9
##           Reference
## Prediction  No  Yes
##           No 5413 967
##           Yes 131 213
## Sensitivity: 0.181
## Specificity: 0.976
## Balanced Accuracy: 0.578

```

```

##
## ##### Group: 10
##           Reference
## Prediction  No  Yes
##           No 5011 1086
##           Yes 138 198
## Sensitivity: 0.154
## Specificity: 0.973
## Balanced Accuracy: 0.564
##
## ##### Group: 11
##           Reference
## Prediction  No  Yes
##           No 3500 799
##           Yes 152 184
## Sensitivity: 0.187
## Specificity: 0.958
## Balanced Accuracy: 0.573
##
## ##### Group: 12
##           Reference
## Prediction  No  Yes
##           No 2411 539
##           Yes 127 125
## Sensitivity: 0.188
## Specificity: 0.95
## Balanced Accuracy: 0.569
##
## ##### Group: 13
##           Reference
## Prediction  No  Yes
##           No 2684 532
##           Yes 188 142
## Sensitivity: 0.211
## Specificity: 0.935
## Balanced Accuracy: 0.573
##
## ### Fairness Analysis by Education
##
## ##### Group: 1
##           Reference
## Prediction No Yes
##           No 23 8
##           Yes 1 3
## Sensitivity: 0.273
## Specificity: 0.958
## Balanced Accuracy: 0.616
##
## ##### Group: 2
##           Reference
## Prediction No Yes
##           No 489 167
##           Yes 61 89
## Sensitivity: 0.348

```

```

## Specificity: 0.889
## Balanced Accuracy: 0.618
##
## ##### Group: 3
##           Reference
## Prediction  No  Yes
##           No 1286 343
##           Yes 116 133
## Sensitivity: 0.279
## Specificity: 0.917
## Balanced Accuracy: 0.598
##
## ##### Group: 4
##           Reference
## Prediction  No  Yes
##           No 10133 1777
##           Yes 346 397
## Sensitivity: 0.183
## Specificity: 0.967
## Balanced Accuracy: 0.575
##
## ##### Group: 5
##           Reference
## Prediction  No  Yes
##           No 11701 1808
##           Yes 254 318
## Sensitivity: 0.15
## Specificity: 0.979
## Balanced Accuracy: 0.564
##
## ##### Group: 6
##           Reference
## Prediction  No  Yes
##           No 19083 1831
##           Yes 173 195
## Sensitivity: 0.096
## Specificity: 0.991
## Balanced Accuracy: 0.544
##
## ### Fairness Analysis by Income
##
## ##### Group: 1
##           Reference
## Prediction  No  Yes
##           No 1372 358
##           Yes 123 173
## Sensitivity: 0.326
## Specificity: 0.918
## Balanced Accuracy: 0.622
##
## ##### Group: 2
##           Reference
## Prediction  No  Yes
##           No 1555 425

```

```

##          Yes  178  171
## Sensitivity: 0.287
## Specificity: 0.897
## Balanced Accuracy: 0.592
##
## ##### Group: 3
##          Reference
## Prediction   No  Yes
##           No 2357 517
##           Yes  138 176
## Sensitivity: 0.254
## Specificity: 0.945
## Balanced Accuracy: 0.599
##
## ##### Group: 4
##          Reference
## Prediction   No  Yes
##           No 3095 673
##           Yes  113 163
## Sensitivity: 0.195
## Specificity: 0.965
## Balanced Accuracy: 0.58
##
## ##### Group: 5
##          Reference
## Prediction   No  Yes
##           No 4226 780
##           Yes  124 155
## Sensitivity: 0.166
## Specificity: 0.971
## Balanced Accuracy: 0.569
##
## ##### Group: 6
##          Reference
## Prediction   No  Yes
##           No 6110 919
##           Yes  113 128
## Sensitivity: 0.122
## Specificity: 0.982
## Balanced Accuracy: 0.552
##
## ##### Group: 7
##          Reference
## Prediction   No  Yes
##           No 7584 922
##           Yes   63  97
## Sensitivity: 0.095
## Specificity: 0.992
## Balanced Accuracy: 0.543
##
## ##### Group: 8
##          Reference
## Prediction   No  Yes
##           No 16416 1340

```

```

##          Yes    99    72
## Sensitivity: 0.051
## Specificity: 0.994
## Balanced Accuracy: 0.522

##
## ### Fairness Analysis by Sex
##
## #### Group: 0
##          Reference
## Prediction    No    Yes
##           No 18017   874
##           Yes  6720  2812
## Sensitivity: 0.763
## Specificity: 0.728
## Balanced Accuracy: 0.746
##
## #### Group: 1
##          Reference
## Prediction    No    Yes
##           No 13228   872
##           Yes  5701  2511
## Sensitivity: 0.742
## Specificity: 0.699
## Balanced Accuracy: 0.721
##
## ### Fairness Analysis by Age
##
## #### Group: 1
##          Reference
## Prediction    No    Yes
##           No  1137   15
##           Yes   12    0
## Sensitivity: 0
## Specificity: 0.99
## Balanced Accuracy: 0.495
##
## #### Group: 2
##          Reference
## Prediction    No    Yes
##           No  1430   21
##           Yes   33    9
## Sensitivity: 0.3
## Specificity: 0.977
## Balanced Accuracy: 0.639
##
## #### Group: 3
##          Reference
## Prediction    No    Yes
##           No  2068   46
##           Yes   75   17
## Sensitivity: 0.27
## Specificity: 0.965
## Balanced Accuracy: 0.617

```

```

##
## ##### Group: 4
##           Reference
## Prediction  No  Yes
##           No 2489  89
##           Yes 150  64
## Sensitivity: 0.418
## Specificity: 0.943
## Balanced Accuracy: 0.681
##
## ##### Group: 5
##           Reference
## Prediction  No  Yes
##           No 2755 104
##           Yes 275  99
## Sensitivity: 0.488
## Specificity: 0.909
## Balanced Accuracy: 0.698
##
## ##### Group: 6
##           Reference
## Prediction  No  Yes
##           No 3098 141
##           Yes 495 208
## Sensitivity: 0.596
## Specificity: 0.862
## Balanced Accuracy: 0.729
##
## ##### Group: 7
##           Reference
## Prediction  No  Yes
##           No 3751 222
##           Yes 842 394
## Sensitivity: 0.64
## Specificity: 0.817
## Balanced Accuracy: 0.728
##
## ##### Group: 8
##           Reference
## Prediction  No  Yes
##           No 3940 241
##           Yes 1361 614
## Sensitivity: 0.718
## Specificity: 0.743
## Balanced Accuracy: 0.731
##
## ##### Group: 9
##           Reference
## Prediction  No  Yes
##           No 3672 268
##           Yes 1872 912
## Sensitivity: 0.773
## Specificity: 0.662
## Balanced Accuracy: 0.718

```



```

##
## ##### Group: 10
##           Reference
## Prediction  No  Yes
##           No 3091 239
##           Yes 2058 1045
## Sensitivity: 0.814
## Specificity: 0.6
## Balanced Accuracy: 0.707
##
## ##### Group: 11
##           Reference
## Prediction  No  Yes
##           No 1784 182
##           Yes 1868 801
## Sensitivity: 0.815
## Specificity: 0.488
## Balanced Accuracy: 0.652
##
## ##### Group: 12
##           Reference
## Prediction  No  Yes
##           No 1070 98
##           Yes 1468 566
## Sensitivity: 0.852
## Specificity: 0.422
## Balanced Accuracy: 0.637
##
## ##### Group: 13
##           Reference
## Prediction  No  Yes
##           No 960 80
##           Yes 1912 594
## Sensitivity: 0.881
## Specificity: 0.334
## Balanced Accuracy: 0.608
##
## ### Fairness Analysis by Education
##
## ##### Group: 1
##           Reference
## Prediction No Yes
##           No 9 0
##           Yes 15 11
## Sensitivity: 1
## Specificity: 0.375
## Balanced Accuracy: 0.688
##
## ##### Group: 2
##           Reference
## Prediction No Yes
##           No 191 30
##           Yes 359 226
## Sensitivity: 0.883

```

```

## Specificity: 0.347
## Balanced Accuracy: 0.615
##
## ##### Group: 3
##           Reference
## Prediction  No  Yes
##           No  637  48
##           Yes 765 428
## Sensitivity: 0.899
## Specificity: 0.454
## Balanced Accuracy: 0.677
##
## ##### Group: 4
##           Reference
## Prediction  No  Yes
##           No  6213 404
##           Yes 4266 1770
## Sensitivity: 0.814
## Specificity: 0.593
## Balanced Accuracy: 0.704
##
## ##### Group: 5
##           Reference
## Prediction  No  Yes
##           No  8456 530
##           Yes 3499 1596
## Sensitivity: 0.751
## Specificity: 0.707
## Balanced Accuracy: 0.729
##
## ##### Group: 6
##           Reference
## Prediction  No  Yes
##           No  15739 734
##           Yes  3517 1292
## Sensitivity: 0.638
## Specificity: 0.817
## Balanced Accuracy: 0.728
##
## ### Fairness Analysis by Income
##
## ##### Group: 1
##           Reference
## Prediction  No  Yes
##           No  685  51
##           Yes 810 480
## Sensitivity: 0.904
## Specificity: 0.458
## Balanced Accuracy: 0.681
##
## ##### Group: 2
##           Reference
## Prediction  No  Yes
##           No  698  64

```

```

##          Yes 1035  532
## Sensitivity: 0.893
## Specificity: 0.403
## Balanced Accuracy: 0.648
##
## ##### Group: 3
##          Reference
## Prediction   No  Yes
##           No 1241  88
##           Yes 1254 605
## Sensitivity: 0.873
## Specificity: 0.497
## Balanced Accuracy: 0.685
##
## ##### Group: 4
##          Reference
## Prediction   No  Yes
##           No 1778 143
##           Yes 1430 693
## Sensitivity: 0.829
## Specificity: 0.554
## Balanced Accuracy: 0.692
##
## ##### Group: 5
##          Reference
## Prediction   No  Yes
##           No 2622 172
##           Yes 1728 763
## Sensitivity: 0.816
## Specificity: 0.603
## Balanced Accuracy: 0.709
##
## ##### Group: 6
##          Reference
## Prediction   No  Yes
##           No 4210 253
##           Yes 2013 794
## Sensitivity: 0.758
## Specificity: 0.677
## Balanced Accuracy: 0.717
##
## ##### Group: 7
##          Reference
## Prediction   No  Yes
##           No 5841 339
##           Yes 1806 680
## Sensitivity: 0.667
## Specificity: 0.764
## Balanced Accuracy: 0.716
##
## ##### Group: 8
##          Reference
## Prediction   No  Yes
##           No 14170 636

```

```
##           Yes  2345   776
## Sensitivity: 0.55
## Specificity: 0.858
## Balanced Accuracy: 0.704
```

```
knitr::opts_chunk$set(echo = FALSE, warning = FALSE, message = FALSE)
library(tidyverse)
library(glmnet)
library(caret)
library(corrplot)
library(pROC)
library(rpart)
library(rpart.plot)
# Load data
data <- read.csv("diabetes_data.csv")

# Convert target to factor
data$Diabetes_binary <- as.factor(data$Diabetes_binary)
ggplot(data, aes(x = Diabetes_binary)) +
  geom_bar(fill = "steelblue") +
  labs(title = "Distribution of Diabetes Status", x = "Diabetes (0 = No, 1 = Yes)", y = "Count") +
  theme_minimal()
# Calculate correlation matrix for numeric predictors
numeric_vars <- data %>% select(-Diabetes_binary)
cor_matrix <- cor(numeric_vars)
corrplot(cor_matrix, method = "color", type = "upper", tl.col = "black", tl.cex = 0.7)
# Find and print pairs with specific correlation strengths
cor_melted <- as.data.frame(as.table(cor_matrix))
names(cor_melted) <- c("Var1", "Var2", "Correlation")

# Filter out self-correlations and duplicate pairs
cor_melted <- cor_melted %>%
  filter(as.character(Var1) < as.character(Var2))

print_correlation_pairs <- function(df, lower_bound, upper_bound, strength_label) {
  pairs <- df %>%
    filter(abs(Correlation) >= lower_bound & abs(Correlation) < upper_bound) %>%
    arrange(desc(abs(Correlation)))

  if (nrow(pairs) > 0) {
    cat(paste0("\n--- ", strength_label, " Correlations (", lower_bound, "-", upper_bound, ") ---\n"))
    for (i in 1:nrow(pairs)) {
      cat(sprintf("%s - %s: %.2f\n", pairs$Var1[i], pairs$Var2[i], pairs$Correlation[i]))
    }
  } else {
    cat(paste0("\n--- No ", strength_label, " Correlations found (", lower_bound, "-", upper_bound, ") \n"))
  }
}

print_correlation_pairs(cor_melted, 0.2, 0.4, "Weak")
print_correlation_pairs(cor_melted, 0.4, 0.6, "Moderate")
print_correlation_pairs(cor_melted, 0.6, 1.0, "Strong")
# Calculate by inverting the correlation matrix
X <- data %>% select(-Diabetes_binary)
```

```

# turn X into a matrix
X <- as.matrix(X)
# calculate the correlation matrix
cor_matrix <- cor(X)
# calculate the VIF
vif <- diag(solve(cor_matrix))

# Visualize VIF
barplot(vif, main = "Variance Inflation Factor", xlab = "Variable", ylab = "VIF", horiz = TRUE)
set.seed(123)
# Ensure target is a factor with valid names for caret
data$Diabetes_binary <- factor(data$Diabetes_binary, levels = c("0", "1"), labels = c("No", "Yes"))

trainIndex <- createDataPartition(data$Diabetes_binary,
  p = .8,
  list = FALSE,
  times = 1
)
dataTrain <- data[trainIndex, ]
dataTest <- data[-trainIndex, ]
# Define train control with 5-fold CV and class probabilities
ctrl <- trainControl(
  method = "cv",
  number = 5,
  summaryFunction = twoClassSummary,
  classProbs = TRUE,
  savePredictions = "final"
)
# Standard Logistic Regression (No sampling)
set.seed(123)
model_glm <- train(Diabetes_binary ~ .,
  data = dataTrain,
  method = "glm",
  family = "binomial",
  metric = "ROC",
  trControl = ctrl
)

# Logistic Regression with Downsampling
ctrl_down <- ctrl
ctrl_down$sampling <- "down"
set.seed(123)
model_glm_down <- train(Diabetes_binary ~ .,
  data = dataTrain,
  method = "glm",
  family = "binomial",
  metric = "ROC",
  trControl = ctrl_down
)

# Logistic Regression with Upsampling
ctrl_up <- ctrl
ctrl_up$sampling <- "up"

```



```

coef_1se <- sort(coef_1se, decreasing = TRUE)
print(coef_1se)
coef_1se <- coef_1se[abs(coef_1se) > 0.1]
print(coef_1se)
formula_lasso_1se <- as.formula(
  paste("Diabetes_binary ~ ", paste(names(coef_1se), collapse = " + "))
)
model_lasso_1se <- train(formula_lasso_1se,
  data = dataTrain,
  method = "glm",
  family = "binomial",
  metric = "ROC",
  trControl = ctrl_down
)
plot_residuals <- function(model, dataTrain, var_name) {
  # Calculate residuals on the full training set
  # We use the downsampled model to predict on the original training data
  probs_train <- predict(model, dataTrain, type = "prob")[, "Yes"]
  y_train_num <- ifelse(dataTrain$Diabetes_binary == "Yes", 1, 0)

  # Calculate Pearson residuals
  # Avoid division by zero if probs are 0 or 1 (add small epsilon)
  epsilon <- 1e-6
  probs_train <- pmax(pmin(probs_train, 1 - epsilon), epsilon)
  residuals_pearson <- (y_train_num - probs_train) / sqrt(probs_train * (1 - probs_train))

  # gut check: residuals should be close to 0 for linear relationships
  cat("Gut Check: Residuals should be close to 0 for linear relationships\n")
  cat("Mean Residual: ", mean(residuals_pearson), "\n")

  # Plot Residuals vs BMI
  # We use a sample of points to avoid overplotting if the dataset is huge
  set.seed(123)
  sample_idx <- sample(1:nrow(dataTrain), min(5000, nrow(dataTrain)))

  plot(dataTrain[[var_name]][sample_idx], residuals_pearson[sample_idx],
    main = paste("Pearson Residuals vs", var_name),
    xlab = var_name, ylab = "Pearson Residuals",
    pch = 20, col = rgb(0, 0, 0, 0.2)
  )
  lines(lowess(dataTrain[[var_name]][sample_idx], residuals_pearson[sample_idx]), col = "red", lwd = 2)
}

plot_deviance_residuals <- function(model, dataTrain, var_name) {
  # Predict probabilities on the full training set
  probs_train <- predict(model, dataTrain, type = "prob")[, "Yes"]
  y_train_num <- ifelse(dataTrain$Diabetes_binary == "Yes", 1, 0)

  # Avoid probabilities exactly 0 or 1
  epsilon <- 1e-6
  probs_train <- pmax(pmin(probs_train, 1 - epsilon), epsilon)

  # Deviance residuals for Bernoulli / binomial

```

```

#  $r_i = \text{sign}(y_i - p_i) * \sqrt{2 * [y_i \log(y_i/p_i) + (1-y_i) \log((1-y_i)/(1-p_i))]}$ 
# Handle  $y_i = 0$  or  $1$  explicitly so we never take  $\log(0)$ 
term1 <- ifelse(y_train_num == 0,
  0,
  y_train_num * log(y_train_num / probs_train)
)
term2 <- ifelse(y_train_num == 1,
  0,
  (1 - y_train_num) * log((1 - y_train_num) / (1 - probs_train))
)

dev_i <- 2 * (term1 + term2)
residuals_deviance <- sign(y_train_num - probs_train) * sqrt(dev_i)

# Gut check
cat("Gut Check: Deviance residuals should be roughly symmetric around 0\n")
cat("Mean Residual: ", mean(residuals_deviance), "\n")

# Plot Residuals vs chosen variable
set.seed(123)
sample_idx <- sample(1:nrow(dataTrain), min(5000, nrow(dataTrain)))

plot(dataTrain[[var_name]][sample_idx], residuals_deviance[sample_idx],
  main = paste("Deviance Residuals vs", var_name),
  xlab = var_name, ylab = "Deviance Residuals",
  pch = 20, col = rgb(0, 0, 0, 0.2)
)
lines(lowess(dataTrain[[var_name]][sample_idx], residuals_deviance[sample_idx]),
  col = "red", lwd = 2
)
}

plot_deviance_residuals(model_glm, dataTrain, "BMI")
# Function to calculate metrics
calc_metrics <- function(model, test_data) {
  probs <- predict(model, test_data, type = "prob")[, "Yes"]
  preds <- predict(model, test_data, type = "raw")

  cm <- confusionMatrix(preds, test_data$Diabetes_binary, positive = "Yes")
  roc_obj <- roc(test_data$Diabetes_binary, probs)
  auc_val <- auc(roc_obj)

  list(
    CM = cm,
    AUC = auc_val,
    F1 = cm$byClass["F1"],
    Recall = cm$byClass["Sensitivity"],
    Precision = cm$byClass["Precision"]
  )
}

metrics_glm <- calc_metrics(model_glm, dataTest)
metrics_down <- calc_metrics(model_glm_down, dataTest)
metrics_up <- calc_metrics(model_glm_up, dataTest)

```



```

metrics_lasso <- calc_metrics(model_lasso, dataTest)
metrics_lasso_1se <- calc_metrics(model_lasso_1se, dataTest)

# Display Comparison
results_df <- data.frame(
  Model = c("GLM (Standard)", "GLM (Downsample)", "GLM (Upsample)", "LASSO (Downsample)", "LASSO (1SE)"),
  AUC = c(metrics_glm$AUC, metrics_down$AUC, metrics_up$AUC, metrics_lasso$AUC, metrics_lasso_1se$AUC),
  Recall = c(metrics_glm$Recall, metrics_down$Recall, metrics_up$Recall, metrics_lasso$Recall, metrics_lasso_1se$Recall),
  F1 = c(metrics_glm$F1, metrics_down$F1, metrics_up$F1, metrics_lasso$F1, metrics_lasso_1se$F1)
)

knitr::kable(results_df, digits = 3, caption = "Model Performance Comparison")

# Display Confusion Matrices for each model
metrics_glm$CM
metrics_down$CM
metrics_up$CM
metrics_lasso$CM
metrics_lasso_1se$CM

# Extract coefficients
plot_importances <- function(model, model_name) {
  coefs <- coef(model$finalModel)
  odds_ratios <- exp(coefs)
  odds_df <- data.frame(Feature = names(odds_ratios), OddsRatio = odds_ratios) %>%
    filter(Feature != "(Intercept)") %>%
    arrange(desc(OddsRatio))

  ggplot(odds_df, aes(
    x = reorder(Feature, OddsRatio),
    y = OddsRatio,
    fill = OddsRatio > 1
  )) +
    geom_col() +
    scale_fill_manual(values = c(`TRUE` = "steelblue", `FALSE` = "red")) +
    coord_flip() +
    labs(
      title = paste0("Feature Importance (Odds Ratios) - ", model_name),
      x = "Feature",
      y = "Multiplicative Impact on Odds",
      fill = "Odds > 1"
    ) +
    theme_minimal()
}

plot_importances(model_glm, "GLM (Standard)")
plot_importances(model_lasso_1se, "GLM (Lasso + 1SE)")
plot_importances(model_glm_down, "GLM (Downsample)")
# Define demographic variables to check
demo_vars <- c("Sex", "Age", "Education", "Income")

# Function to print confusion matrix for subgroups
analyze_fairness <- function(var_name, model, data) {
  cat(paste0("\n### Fairness Analysis by ", var_name, "\n"))

```

```

groups <- unique(data[[var_name]])

for (g in sort(groups)) {
  sub_data <- data[data[[var_name]] == g, ]
  if (nrow(sub_data) > 0) {
    preds <- predict(model, sub_data)
    cm <- confusionMatrix(preds, sub_data$Diabetes_binary, positive = "Yes")
    cat(paste0("\n#### Group: ", g, "\n"))
    print(cm$table)
    cat(paste0("Sensitivity: ", round(cm$byClass["Sensitivity"], 3), "\n"))
    cat(paste0("Specificity: ", round(cm$byClass["Specificity"], 3), "\n"))
    cat(paste0("Balanced Accuracy: ", round(cm$byClass["Balanced Accuracy"], 3), "\n"))
  }
}

# Run fairness analysis
for (var in demo_vars) {
  if (var %in% names(dataTest)) {
    analyze_fairness(var, model_glm_down, dataTest)
  }
}

for (var in demo_vars) {
  if (var %in% names(dataTest)) {
    analyze_fairness(var, model_glm, dataTest)
  }
}

for (var in demo_vars) {
  if (var %in% names(dataTest)) {
    analyze_fairness(var, model_lasso_1se, dataTest)
  }
}

```